Example: Let $X = \mathbb{R}$ & $\mathcal{B} = \{(a, b] \mid a, b \in \mathbb{R}, a < b\}$. Show that $\mathcal{B}$ is a base for some topology τ on $\mathbb{R}$. τ is called the upper limit topology on $\mathbb{R}$.

Sol^n We shall show that $\mathcal{B}$ satisfies (B1) & (B2).

(B1) Let $x \in \mathbb{R}$. Then $x \in (x-1, x]$.

Thus

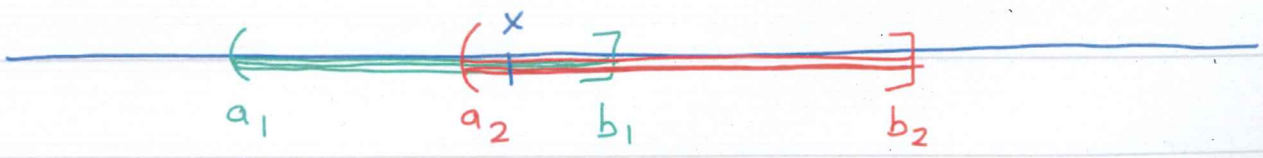
$$\mathbb{R} \subseteq \bigcup_{x \in \mathbb{R}} (x-1, x] = \bigcup \mathcal{B}' \subseteq \bigcup \mathcal{B} \subseteq \mathbb{R}$$

where $\mathcal{B}' = \{(x-1, x] \mid x \in \mathbb{R}\}$.

Therefore

$$\bigcup \mathcal{B} = \mathbb{R}.$$

(B2) Let $B_1 = (a_1, b_1]$, $B_2 = (a_2, b_2] \in \mathcal{B}$ & $x \in (a_1, b_1] \cap (a_2, b_1] = B_1 \cap B_2$.



Set $a_3 = \max\{a_1, a_2\}$ & $b_3 = \min\{b_1, b_2\}$

Let $B_3 = (a_3, b_3]$. Then

$x \in B_3 \subseteq B_1 \cap B_2$.

$\therefore$ $\mathcal{B}$ is a base for some top. τ on $\mathbb{R}$.

Example: Let $X = \mathbb{R}^2$ & let $\mathcal{B}$ be the class of all open discs in $\mathbb{R}^2$. Show that $\mathcal{B}$ is a base for some top. τ on $\mathbb{R}^2$. (τ is the usual top. on $\mathbb{R}^2$).

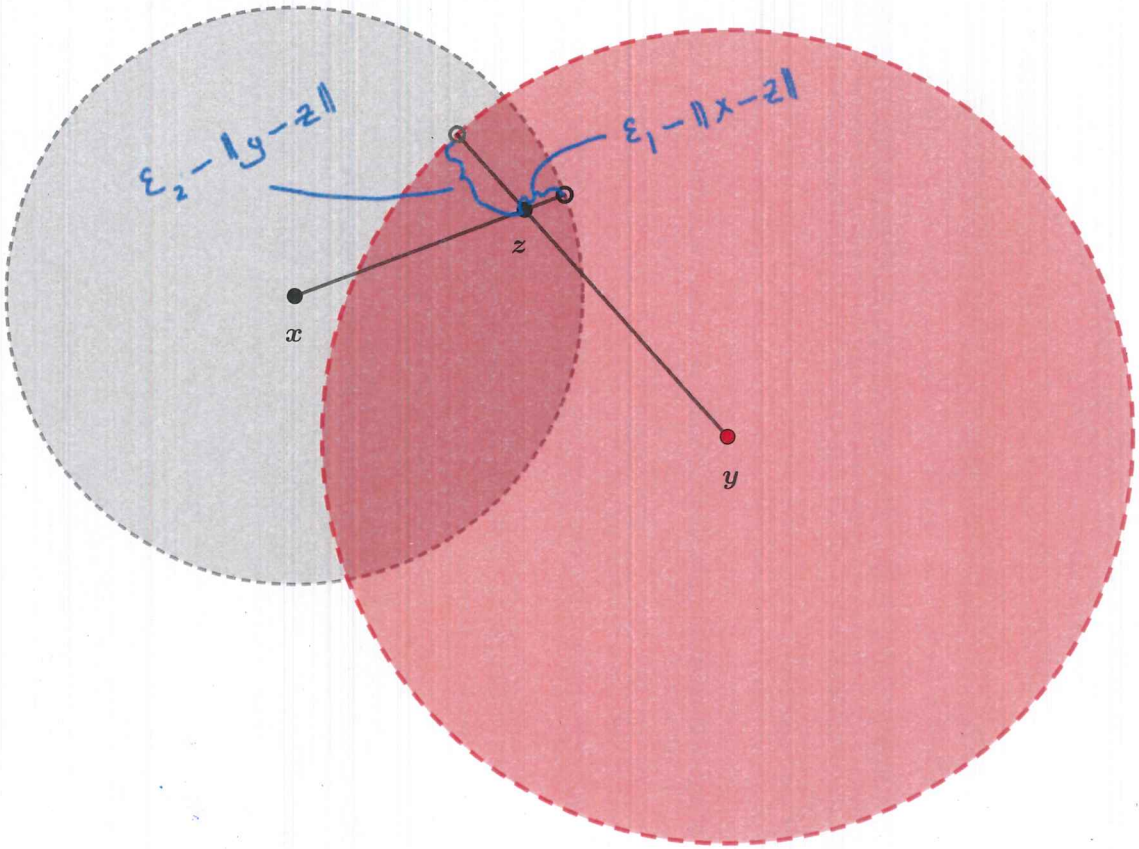
Sol^ As in the previous example we shall show that $\mathcal{B}$ possesses (B1) & (B2).

(B1) Let $x = (x_1, x_2) \in \mathbb{R}^2$. Then $x \in B(x, 1)$. Hence if $\mathcal{B}' = \{ B(x, 1) \mid x \in \mathbb{R}^2 \}$ we have

$$\mathbb{R}^2 \subseteq \bigcup_{x \in \mathbb{R}^2} B(x,1) = \bigcup \mathcal{B}' \subseteq \bigcup \mathcal{B} \subseteq \mathbb{R}^2$$

$$\therefore \quad \bigcup \mathcal{B} = \mathbb{R}^2.$$

(B2). Let $B_1 = B(x, \varepsilon_1)$, $B_2 = B(y, \varepsilon_2) \in \mathcal{B}$ & let $z \in B_1 \cap B_2$.



Let $\varepsilon > 0$ be chosen in such a way that

$$0 < \varepsilon < \min\left\{\varepsilon_1 - \|x-z\|, \varepsilon_2 - \|y-z\|\right\} \leq \varepsilon_2 - \|y-z\|$$

We claim that $B_3 = B(z, \varepsilon) \subseteq B_1 \cap B_2$.

Let $w \in B_3 = B(z, \varepsilon)$.

$$\|x - w\| \leq \|x - z\| + \|z - w\|$$
$$< (\varepsilon_1 - \varepsilon) + \varepsilon = \varepsilon_1$$

$\Rightarrow w \in B(x, \varepsilon_1) = B_1$.

Similarly

$$\|y - w\| \leq \|y - z\| + \|z - w\|$$
$$< (\varepsilon_2 - \varepsilon) + \varepsilon = \varepsilon_2$$

$\Rightarrow w \in B(y, \varepsilon_2) = B_2$.

$\therefore w \in B_3 \Rightarrow w \in B_1 \cap B_2$, i.e.

$z \in B_3 \subseteq B_1 \cap B_2$.

$\therefore \mathcal{B}$ is a base for the usual top. on $\mathbb{R}^2$.

Example: Let $X = \mathbb{R}$ &

$$\mathcal{B} = \{ [p,q] \mid p,q \in \mathbb{Q}, p < q \}.$$

Determine if $\mathcal{B}$ is a base for any topology on $\mathbb{R}$.

Sol^n

Let $B_1 = [1,2]$, $B_2 = [2,3] \in \mathcal{B}$.

Then $2 \in B_1 \cap B_2 = \{2\}$. But we cannot find rational numbers p, q with $p < q$ satisfying

$2 \in B_3 = [p,q] \subseteq B_1 \cap B_2 = \{2\}$.

Thus $\mathcal{B}$ cannot be a base for any topology on $\mathbb{R}$.

Example: Let $\mathcal{B}$ be a base for a topology $\mathcal{T}$ on a set X & let $\mathcal{B}'$ be a class of subsets of X satisfying $\mathcal{B} \subseteq \mathcal{B}' \subseteq \mathcal{T}$. Show that $\mathcal{B}'$ is also a base for $\mathcal{T}$.

Sol^n Let $U \in \mathcal{T}$ & $x \in U$. Since $\mathcal{B}$ is a base for $\mathcal{T}$ there exists a $B \in \mathcal{B}$ s.t. $x \in B \subseteq U$.

Because $\mathcal{B} \subseteq \mathcal{B}'$, $B \in \mathcal{B}'$, i.e. for every $U \in \mathcal{T}$ & every $x \in U$ there exists a $B \in \mathcal{B}'$ s.t. $x \in B \subseteq U$.

$\therefore$ $\mathcal{B}'$ is a base for $\mathcal{T}$.

Example: Let $(X, \mathcal{T}), (X, \vartheta)$ be topological spaces. Suppose that $\mathcal{B}$ & η are bases for $\mathcal{T}$ & ϑ, resp. Show that $\vartheta \subseteq \mathcal{T} \iff$ for every $N \in \eta$ & every $x \in N$ there exists a $B \in \mathcal{B}$ s.t. $x \in B \subseteq N$.

Sol^n

$\Rightarrow$ Assume that $\vartheta \subseteq \tau$. Let $N \in \eta$ & $x \in N$. Since $\eta \subseteq \vartheta \subseteq \tau$, $N \in \tau$ & $x \in N$. Because $\mathcal{B}$ is a base for τ there exists a $B \in \mathcal{B}$ s.t. $x \in B \subseteq N$.

$\Leftarrow$ Assume that for every $N \in \eta$ & every $x \in N$ there exists a $B \in \mathcal{B}$ s.t. $x \in B \subseteq N$. Let $V \in \vartheta$.

Case 1 If $V = \emptyset$ then $V \in \tau$.

Case 2 Assume that $V \neq \emptyset$. Then there exists an $x \in X$ s.t. $x \in V$. Since η is a base for ϑ there exists an $N \in \eta$ s.t. $x \in N \subseteq V$. By our assumption there exists a $B \in \mathcal{B}$ s.t. $x \in B \subseteq N \subseteq V$, i.e. for every $V \in \vartheta$ & $x \in V$ there exists a $B \in \mathcal{B}$ s.t. $x \in B \subseteq V$. But $\mathcal{B}$ is a base for τ. Thus $V \in \tau$, i.e. $\vartheta \subseteq \tau$.

Example: Let $(\mathbb{R}, \mathcal{U})$ & $(\mathbb{R}, \mathcal{T})$ be the usual topological space & the upper limit topological space, resp. Show that $\mathcal{U} \subseteq \mathcal{T}$.

Soln

We know that

$\eta = \{(a,b) \mid a,b \in \mathbb{R}, a < b\}$ &

$\mathcal{B} = \{(a,b] \mid a,b \in \mathbb{R}, a < b\}$

are bases for $\mathcal{U}$ & $\mathcal{T}$, resp.

We shall use the previous example to show that $\mathcal{U} \subseteq \mathcal{T}$. If we show that for every $N \in \eta$ & every $x \in N$ there exists a $B \in \mathcal{B}$ s.t. $x \in B \subseteq N$ we are done.

Let $N = (a,b)$ & $x \in N = (a,b)$.

We know that

$$(a,b) = \bigcup_{n=1}^{\infty} \left(a, b - \frac{1}{n}\right]$$

& $x \in N = (a, b)$. Therefore there exists an $n_0 \in \mathbb{Z}^+$ s.t.

$$x \in \underbrace{\left(a, b - \frac{1}{n_0}\right]}_{B \in \mathcal{B}} \subseteq (a, b) = N.$$

$\therefore \quad \mathcal{T} \subseteq \tau.$